

SOME ELECTROMAGNETIC FIELDS OF CYLINDRICAL SYMMETRY

by MARKANDEYA MISRA and L. RADHAKRISHNA, *Department of Mathematics,
Banaras Hindu University, Varanasi 5*

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ABSTRACT

The non-static cylindrically symmetric metric of Einstein and Rosen is considered and several exact solutions of the Einstein-Maxwell field equations *in vacuo* are obtained, some of which represent null electromagnetic fields. It is found that the metric in case of the null electromagnetic fields is plane symmetric in the sense of Taub and consequently the corresponding solutions represent plane electromagnetic waves. A method is obtained which enables one to construct solutions, under certain conditions, representing electromagnetic fields *in vacuo* from known solutions of the empty space-time field equations and vice versa. Certain results, similar to those obtained by Weyl in the static axially symmetric case, are also obtained.

1. INTRODUCTION

The metric

$$(1) \quad ds^2 = e^{\lambda - \mu} (dt^2 - d\rho^2) - \rho^2 e^{-\mu} d\Phi^2 - e^{\mu} dz^2,$$

λ and μ being functions of ρ and t only, has been the subject of extensive investigation by Einstein and Rosen (1937), Rosen (1954, 1955, 1958), Weber and Wheeler (1957), Bonnor (1957) and others in connection with gravitational radiation in empty space-time. According to Pirani's criterion (1957), when $R_{\alpha\beta} = 0$, it is of type II with radiation present if a certain function of λ and μ be non-zero.

There does not, however, seem to have been any attempt so far to obtain solutions corresponding to (1) which represent an electromagnetic energy distribution. As regards perfect fluid distributions, it can easily be shown that two of the eigenvalues of T^β_α with respect to g^β_α corresponding to (1) are equal and opposite. Consequently this possibility is ruled out. Accordingly we concern ourselves here with the other possibility, i.e. to obtain solutions of the Einstein-Maxwell field equations *in vacuo*.

Weyl (1917), in the static axially symmetric case, showed that the most general relation which can exist between g_{44} and A_4 (A_4 being the only non-zero component of the 4-potential A_x in the electrostatic case considered by him) is of the form

$$(A) \quad g_{44} = \alpha + \beta A_4 + 4\pi (A_4)^2,$$

α, β being arbitrary constants. This result was later extended by Majumdar (1947) to the more general static metric

$$ds^2 = g_{44}dt^2 + g_{ij}dx^i dx^j, \quad (i, j = 1, 2, 3),$$

$g_{\alpha\beta}$ being functions of x^1, x^2 and x^3 only. In the non-static cylindrically symmetric case, however, a relationship similar to (A) has not been obtained. In the present investigation we have shown that the most general relation which can exist between g_{33} and A_3 (A_3 being the only non-zero component of the 4-potential A_α , considered here as a special case) is of the form

$$(B) \quad g_{33} = -\alpha - \beta A_3 + 4\pi(A_3)^2,$$

α, β being arbitrary constants. A similar relation is also established between g_{22} and A_2 if only A_2 be non-zero.

In the case when (B) holds it is shown here that corresponding to every set of g_{ij}, F_{ij} which satisfy the Einstein-Maxwell field equations *in vacuo* there exist ${}^*g_{ij}$ which are connected with g_{ij} and which satisfy the field equations for empty space-time and vice versa. Thus knowing a solution of the empty space-time field equations we can construct a solution which represents an electromagnetic field *in vacuo* for which the relation (B) holds and conversely.

Several exact solutions of the Einstein-Maxwell field equations *in vacuo* are obtained, many of which represent null electromagnetic fields. It is found that the solutions corresponding to null electromagnetic fields as obtained here are plane symmetric in the sense of Taub (1951).

2. THE FIELD EQUATIONS AND SOME GENERAL CONSIDERATIONS

The field equations for regions containing an electromagnetic field but no matter are:

$$(2) \quad R_\alpha^\beta = -8\pi E_\alpha^\beta,$$

and

$$(3) \quad (\sqrt{-g}F^{\mu\nu})_{,\nu} = 0,$$

where

$$(4) \quad E_\alpha^\beta = -F_{\mu\alpha}F^{\mu\beta} + \frac{1}{4}g_\alpha^\beta F_{\sigma\epsilon}F^{\sigma\epsilon}$$

and

$$(5) \quad F_{\mu\nu} = A_{\mu,\nu} - A_{\nu,\mu}.$$

Here R_α^β is the contracted curvature tensor, E_α^β the electromagnetic energy tensor, $F_{\alpha\beta}$ the electromagnetic field tensor and A_α the 4-potential. A comma followed by a lower suffix denotes a partial differentiation with respect to the corresponding variable. The charge-current density has been put equal to zero.

The non-zero components of the Ricci tensor for the metric (1) are

$$(6) \quad \begin{cases} R_1^1 = \frac{1}{2}e^{\mu-\lambda} \left(-\lambda_{11} + \lambda_{44} + \mu_{11} - \mu_{44} - \mu_1^2 + \frac{\lambda_1 + \mu_1}{\rho} \right), \\ R_2^2 = -R_3^3 = \frac{1}{2}e^{\mu-\lambda} \left(\mu_{11} - \mu_{44} + \frac{\mu_1}{\rho} \right), \\ R_4^4 = \frac{1}{2}e^{\mu-\lambda} \left(-\lambda_{11} + \lambda_{44} + \mu_{11} - \mu_{44} + \mu_4^2 + \frac{\mu_1 - \lambda_1}{\rho} \right), \\ R_1^4 = -R_4^1 = \frac{1}{2}e^{\mu-\lambda} \left(\mu_1 \mu_4 - \frac{\lambda_4}{\rho} \right). \end{cases}$$

The lower suffixes 1 and 4 after an unknown function denote a partial differentiation with respect to ρ and t respectively. Also ρ , Φ , z and t correspond respectively to x^1 , x^2 , x^3 and x^4 .

From (6) we have

$$R_2^2 + R_3^3 = 0,$$

and consequently from (2) and (4) it follows that

$$E_2^2 + E_3^3 = g^{11}g^{44}(F_{14})^2 - g^{22}g^{33}(F_{23})^2 = 0.$$

Since g^{11} , g^{22} and g^{33} are all negative and g^{44} positive it is clear that the above equation will hold only if

$$(7) \quad F_{14} = F_{23} = 0.$$

The remaining components of $F_{\alpha\beta}$ can, by virtue of (5), be derived from only two components of the 4-potential, namely A_2 and A_3 . For convenience we put

$$(8) \quad A_2 = \frac{\phi}{\sqrt{8\pi}}, \quad A_3 = \frac{\psi}{\sqrt{8\pi}}$$

so that

$$(9) \quad F_{12} = -\frac{\phi_1}{\sqrt{8\pi}}, \quad F_{13} = -\frac{\psi_1}{\sqrt{8\pi}}, \quad F_{24} = \frac{\phi_4}{\sqrt{8\pi}}, \quad F_{34} = \frac{\psi_4}{\sqrt{8\pi}}.$$

From (4), (7) and (9) the nonvanishing independent components of the electromagnetic energy tensor are found to be

$$(10) \quad \begin{cases} E_4^4 = -E_1^1 = \frac{1}{16\pi} e^{-\lambda} \left[\frac{1}{\rho^2} e^{2\mu} (\phi_1^2 + \phi_4^2) + \psi_1^2 + \psi_4^2 \right], \\ E_2^2 = -E_3^3 = \frac{1}{16\pi} e^{-\lambda} \left[-\frac{1}{\rho^2} e^{2\mu} (\phi_1^2 - \phi_4^2) + \psi_1^2 - \psi_4^2 \right], \\ E_1^4 = \frac{1}{8\pi} e^{-\lambda} \left[\frac{1}{\rho^2} e^{2\mu} \phi_1 \phi_4 + \psi_1 \psi_4 \right], \\ E_2^3 = \frac{1}{8\pi} e^{-\lambda} \left[-\phi_1 \psi_1 + \phi_4 \psi_4 \right]. \end{cases}$$

Now the field equations (2) with the help of (6) and (10) can be written as:

$$(11) \quad \lambda_{11} - \lambda_{44} + \frac{1}{2}(\mu_1^2 - \mu_4^2) = \frac{e^\mu}{\rho^2}(\phi_1^2 - \phi_4^2) - e^{-\mu}(\psi_1^2 - \psi_4^2),$$

$$(12) \quad \mu_{11} - \mu_{44} + \frac{\mu_1}{\rho} = \frac{e^\mu}{\rho^2}(\phi_1^2 - \phi_4^2) - e^{-\mu}(\psi_1^2 - \psi_4^2),$$

$$(13) \quad \lambda_1 = \frac{1}{2}\rho(\mu_1^2 + \mu_4^2) + \frac{e^\mu}{\rho}(\phi_1^2 + \phi_4^2) + \rho e^{-\mu}(\psi_1^2 + \psi_4^2),$$

$$(14) \quad \lambda_4 = \rho\mu_1\mu_4 + \frac{2}{\rho}e^\mu\phi_1\phi_4 + 2\rho e^{-\mu}\psi_1\psi_4,$$

$$(15) \quad \phi_1\psi_1 - \phi_4\psi_4 = 0.$$

Also from (3) and (9) we get

$$(16) \quad \phi_{11} - \phi_{44} - \frac{\phi_1}{\rho} = -\mu_1\phi_1 + \mu_4\phi_4,$$

$$(17) \quad \psi_{11} - \psi_{44} + \frac{\psi_1}{\rho} = \mu_1\psi_1 - \mu_4\psi_4.$$

The field equations which determine λ , μ , ϕ and ψ are (11), (12), (13), (14), (15), (16) and (17).

It can now easily be shown that the condition of integrability of (13) and (14), i.e.

$$\lambda_{14} = \lambda_{41}$$

is satisfied if (12), (16) and (17) hold. Also (11) is identically satisfied, if (12), (13), (14), (16) and (17) hold. Hence the problem of solving equations (11) to (17) reduces to that of determining μ , ϕ and ψ from (12), (15), (16) and (17), and then λ will be given by (13) and (14).

In the following we attempt to obtain some exact solutions of these field equations.

3. SOLUTIONS OF THE FIELD EQUATIONS

In view of the nonlinear character of the expressions involved, it is extremely difficult to find exact solutions of the field equations. As such we shall make certain simplifying assumptions, and then we obtain the corresponding exact solutions.

We consider solutions of the field equations in the following three cases:

- (i) $\phi = 0$,
- (ii) $\psi = 0$,
- (iii) $\phi \neq 0$, $\psi \neq 0$.

Case (i): $\phi = 0$. The equations (15) and (16) are automatically satisfied and the remaining equations which determine μ , ψ and λ are:

$$(18) \quad \mu_{11} - \mu_{44} + \frac{\mu_1}{\rho} = -e^{-\mu}(\psi_1^2 - \psi_4^2),$$

$$(19) \quad \psi_{11} - \psi_{44} + \frac{\psi_1}{\rho} = \mu_1 \psi_1 - \mu_4 \psi_4,$$

$$(20) \quad \lambda_1 = \frac{1}{2}\rho(\mu_1^2 + \mu_4^2) + \rho e^{-\mu}(\psi_1^2 + \psi_4^2),$$

and

$$(21) \quad \lambda_4 = \rho\mu_1\mu_4 + 2\rho e^{-\mu}\psi_1\psi_4.$$

(i A). *The most general relation between g_{33} and ψ .* We shall first show that the most general relation which can exist between g_{33} and ψ is of the form

$$(22) \quad g_{33} = -\alpha - \beta\psi + \frac{1}{2}\psi^2,$$

α, β being arbitrary constants. In order to show this we assume that μ is a function of ψ which implies that the level surfaces of μ and ψ coincide. Since $\mu_1 = \dot{\mu}\psi_1$, $\mu_{11} = \ddot{\mu}\psi_1^2 + \dot{\mu}\psi_{11}$, etc., where a dot denotes ordinary differentiation with respect to ψ , we find, from (18) and (19), that

$$(23) \quad \ddot{\mu}(\psi_1^2 - \psi_4^2) + \dot{\mu}\left(\psi_{11} - \psi_{44} + \frac{\psi_1}{\rho}\right) = -e^{-\mu}(\psi_1^2 - \psi_4^2),$$

and

$$(24) \quad \psi_{11} - \psi_{44} + \frac{\psi_1}{\rho} = \dot{\mu}(\psi_1^2 - \psi_4^2).$$

Eliminating ψ from (23) and (24) we find that

$$(25) \quad \ddot{\mu} + \dot{\mu}^2 = -e^{-\mu},$$

which leads to the general solution

$$(26) \quad e^{\mu} = \alpha + \beta\psi - \frac{1}{2}\psi^2,$$

α, β being arbitrary constants. Since $g_{33} = -e^{\mu}$ the required result follows.

(i B). In view of (26) equations (18) and (19) become identical and reduce to

$$(27) \quad \psi_{11} - \psi_{44} + \frac{\psi_1}{\rho} = \frac{(\beta - \psi)(\psi_1^2 - \psi_4^2)}{(\alpha + \beta\psi - \frac{1}{2}\psi^2)}.$$

If now we determine ψ from (27), μ will be given by (26) and finally λ can be determined from (20) and (21). That will be the most general solution of the equations (18), (19), (20) and (21) with the relation (26) between μ and ψ .

Consider now the substitution

$$(28) \quad \int \frac{d\psi}{(\alpha + \beta\psi - \frac{1}{2}\psi^2)} = \frac{2x}{\sqrt{2\alpha + \beta^2}},$$

x being a new variable. A simple calculation shows that x satisfies the wave equation

$$(29) \quad x_{11} - x_{44} + \frac{x_1}{\rho} = 0,$$

in view of (27). Hence the problem of solving (27) reduces to that of determining ψ from (28), it being assumed that x is known from (29). Now since g_{33} is negative, it follows from (26) that $\alpha > 0$ and if α is to admit the value zero then obviously $\beta > 0$. Also α and β cannot both vanish simultaneously since that will make g_{33} positive.

Now since $\alpha > 0$ we can write

$$\alpha + \beta\psi - \frac{1}{2}\psi^2 = \frac{1}{2}(p - \psi)(q + \psi),$$

where

$$p - q = 2\beta, \quad pq = 2\alpha,$$

so that

$$p + q = \pm 2\sqrt{2\alpha + \beta^2}.$$

Taking the positive sign we find that

$$p = \beta + \sqrt{2\alpha + \beta^2}, \quad q = -\beta + \sqrt{2\alpha + \beta^2}.$$

Using these in (28) we find that

$$\log \left(\frac{q + \psi}{p - \psi} \right) = 2x,$$

i.e.

$$(30) \quad \psi = \beta + \sqrt{2\alpha + \beta^2} \tanh x.$$

Also equations (20) and (21) in view of (26) and (28) reduce to

$$(31) \quad \lambda_1 = 2\rho(x_1^2 + x_4^2),$$

$$(32) \quad \lambda_4 = 4\rho x_1 x_4.$$

The condition of integrability of (31) and (32) is precisely (29). Hence whenever (29) is satisfied ψ is known from (30) and λ from (31) and (32). Finally from (26) and (30) μ is given by

$$(33) \quad e^\mu = (\alpha + \frac{1}{2}\beta^2) \operatorname{sech}^2 x.$$

Hence we know ψ , λ and μ in terms of x alone, which satisfies the wave equation (29).

Equation (29) is known to have solutions of the form

$$[AJ_0(k\rho) + BY_0(k\rho)] \cos(kt + \epsilon),$$

where A , B , k and ϵ are constants and $J_0(k\rho)$ and $Y_0(k\rho)$ are Bessel's functions of order zero and of the first and second kind respectively. Since $Y_0(k\rho) \rightarrow -\infty$ as $\rho \rightarrow 0$ we take $B = 0$ and a typical solution assumes the form

$$(34) \quad AJ_0(k\rho) \cos(kt + \epsilon).$$

The general solution of (29) can obviously be obtained by superposing terms of the form (34) and from that the solution corresponding to any desired situation can be obtained by a proper choice of the arbitrary constants.

Consider now the metric

$$(35) \quad ds^2 = e^{\lambda-2x}(dt^2 - d\rho^2) - \rho^2 e^{-2x} d\Phi^2 - e^{2x} dz^2,$$

λ and x being functions of ρ and t only. The field equations for empty space-time corresponding to (35) reduce to (29), (31) and (32) (cf. Bonnor 1957). Hence of the five equations (29), (30), (31), (32) and (33), which together with (1) represent an electromagnetic field *in vacuo*, three are the same which along with (35) represent an empty gravitational field. We have therefore obtained the result:

For every solution of (29), (31) and (32) corresponding to (35) which represents empty space-time, we have a solution given by (1), (30) and (33), λ and x remaining the same, which represents an electromagnetic field *in vacuo*.

The converse of the above result is obvious.

Hereafter a, b, c, m, n, p, q appear as arbitrary constants distinct for each solution.

(i C). *Some more exact solutions.* If now we assume that ψ is a function of ρ only then (19) gives the first integral

$$(36) \quad \mu = \log(\rho\psi_1) + F(t),$$

F being an arbitrary function of t only. Substituting for μ from (36) in (18) we find that

$$(37) \quad \frac{d^2}{d\rho^2} \log(\rho\psi_1) + \frac{1}{\rho} \frac{d}{d\rho} \log(\rho\psi_1) - F_{44} = -\frac{\psi_1}{\rho} e^{-F}.$$

Since ψ is a function of ρ only and F a function of t only it follows that (37) will hold only if F be a constant or ψ_1/ρ be a constant. If F is a constant, say

$$(38) \quad F = -\log a,$$

then (37) gives

$$\frac{d^2}{d\rho^2} \log(\rho\psi_1) + \frac{1}{\rho} \frac{d}{d\rho} \log(\rho\psi_1) = -\frac{a\psi_1}{\rho},$$

whose general solution is

$$(39) \quad \psi = p - \frac{m}{a} \tanh\left(-\frac{m}{2} \log \rho + n\right).$$

From (36), (38) and (39) it follows that

$$(40) \quad \mu = \log \left[\frac{m^2}{2a^2} \operatorname{sech}^2 \left(-\frac{m}{2} \log \rho + n \right) \right],$$

and (20), (21), (39) and (40) yield

$$(41) \quad \lambda = \frac{m^2}{2} \log \rho + q.$$

This solution obviously satisfies (30) and (33) if we put

$$\alpha + \frac{\beta^2}{2} = \frac{m^2}{2a^2}, \quad \beta = p \quad \text{and} \quad x = -\frac{m}{2} \log \rho + n.$$

This value of x satisfies (29) as well. Hence substituting for x in (31) and (32) we find that

$$\lambda = \frac{m^2}{2} \log \rho + \text{const.}$$

which is the same as (41). Hence the metric giving empty space-time, associated with the above solution, is

$$ds^2 = \left(\frac{\rho}{\rho_0}\right)^{\frac{m^2+2m}{2}} (dt^2 - d\rho^2) - \left(\frac{\rho}{\rho_0}\right)^{m+2} d\Phi^2 - \left(\frac{\rho}{\rho_0}\right)^{-m} dz^2.$$

This is the solution obtained by Levi-Civita (1919) corresponding to a line-mass placed along the axis of Z .

In the alternative case we put $\psi_1 = a\rho$ in (37), a being an arbitrary constant. The corresponding solution is easily found to be

$$(42) \quad \begin{cases} \psi = \frac{1}{2}a\rho^2 + b, \\ \mu = \log \left[\frac{2a^2}{m^2} \rho^2 \cosh^2 \left(\frac{m}{2} t + n \right) \right], \\ \lambda = \frac{m^2 \rho^2}{4} + \log \left[\rho^2 \cosh^4 \left(\frac{m}{2} t + n \right) \right] + p. \end{cases}$$

If we treat ψ as a function of t only the two solutions in this case are:

$$(43) \quad \begin{cases} \psi = p - \frac{m}{a} \tanh \left(-\frac{m}{2} t + n \right), \\ \mu = \log \left[\frac{m^2}{2a^2} \text{sech}^2 \left(-\frac{m}{2} t + n \right) \right], \\ \lambda = \frac{m^2 \rho^2}{4} + q; \end{cases}$$

and

$$(44) \quad \begin{cases} \psi = at + b, \\ \mu = \log \left[\frac{2a^2}{m^2} \rho^2 \cosh^2 \left(\frac{m}{2} \log \rho + n \right) \right], \\ \lambda = \frac{m^2}{2} \log \rho + \log \left[\rho^2 \cosh^4 \left(\frac{m}{2} \log \rho + n \right) \right] + p. \end{cases}$$

The solution given by (43) satisfies (30) and (33) if we put

$$\alpha + \frac{\beta^2}{2} = \frac{m^2}{2a^2}, \quad \beta = p \text{ and } x = -\frac{m}{2}t + n.$$

(29) is also satisfied. Substituting for x in (31) and (32) λ is given by

$$\lambda = \frac{m^2\rho^2}{4} + \text{const.}$$

which is the same as in (43). Hence the corresponding solution representing empty space-time is given by

$$(45) \quad ds^2 = e^{\frac{m^2\rho^2}{4} + mt} (dt^2 - d\rho^2) - \rho^2 e^{mt} d\Phi^2 - e^{-mt} dz^2.$$

This is a nonstatic solution of the empty space-time field equations and is of type II according to Pirani's criterion (1957).

Case (ii). $\psi = 0$. Equations (15) and (17) are automatically satisfied and the other equations which determine μ , ϕ and λ are:

$$(46) \quad \mu_{11} - \mu_{44} + \frac{\mu_1}{\rho} = \frac{e^\mu}{\rho^2} (\phi_1^2 - \phi_4^2),$$

$$(47) \quad \phi_{11} - \phi_{44} - \frac{\phi_1}{\rho} = -\mu_1 \phi_1 + \mu_4 \phi_4,$$

$$(48) \quad \lambda_1 = \frac{1}{2} \rho (\mu_1^2 + \mu_4^2) + \frac{e^\mu}{\rho} (\phi_1^2 + \phi_4^2),$$

$$(49) \quad \lambda_4 = \rho \mu_1 \mu_4 + \frac{2e^\mu}{\rho} \phi_1 \phi_4.$$

(ii A). It can easily be shown that the most general relation which can exist between g_{22} and ϕ is

$$(50) \quad g_{22} = -\alpha - \beta\phi + \frac{1}{2}\phi^2,$$

α, β being arbitrary constants. This relation is similar to (22).

In order to prove it we proceed as follows:

Let

$$(51) \quad \rho^2 e^{-\mu} = v,$$

so that

$$\mu = 2 \log \rho - \log v.$$

Substituting for μ in (46) and (47) we find that

$$(52) \quad -\frac{1}{v} \left(v_{11} - v_{44} + \frac{v_1}{\rho} \right) + \frac{v_1^2 - v_4^2}{v^2} = \frac{1}{v} (\phi_1^2 - \phi_4^2),$$

$$(53) \quad \phi_{11} - \phi_{44} + \frac{\phi_1}{\rho} = \frac{v_1 \phi_1 - v_4 \phi_4}{v}.$$

If now we assume that v is a function of ϕ we find from (52) and (53) that

$$(54) \quad -\frac{\ddot{v}}{v} (\phi_1^2 - \phi_4^2) - \frac{\dot{v}}{v} \left(\phi_{11} - \phi_{44} + \frac{\phi_1}{\rho} \right) + \frac{\dot{v}^2}{v^2} (\phi_1^2 - \phi_4^2) = \frac{\phi_1^2 - \phi_4^2}{v},$$

$$(55) \quad \phi_{11} - \phi_{44} + \frac{\dot{\phi}_1}{\rho} = \frac{\dot{v}}{v} (\phi_1^2 - \phi_4^2),$$

where a dot denotes ordinary differentiation with respect to ϕ .

Eliminating ϕ from these equations we find that

$$\ddot{v} = -1,$$

whose most general solution is

$$v = \alpha + \beta\phi - \frac{1}{2}\phi^2.$$

Since $g_{22} = -v$ the desired result follows.

(ii B). *Solutions of the field equations in this case.* Now we shall obtain some exact solutions of the equations (46), (47), (48) and (49).

If ϕ be a function of ρ only a procedure similar to that of case (i C) leads to the following two solutions:

$$(56) \quad \begin{cases} \phi = p - \frac{m}{a} \tanh \left(-\frac{m}{2} \log \rho + n \right), \\ \mu = \log \left[\frac{2a^2}{m^2} \rho^2 \cosh^2 \left(-\frac{m}{2} \log \rho + n \right) \right], \\ \lambda = \frac{m^2}{2} \log \rho + \log \left[\rho^2 \cosh^4 \left(-\frac{m}{2} \log \rho + n \right) \right] + q; \end{cases}$$

and

$$(57) \quad \begin{cases} \phi = \frac{1}{2}a\rho^2 + b, \\ \mu = \log \left[\frac{m^2}{2a^2} \operatorname{sech}^2 \left(-\frac{m}{2} t + n \right) \right], \\ \lambda = \frac{m^2\rho^2}{4} + p. \end{cases}$$

If, however, ϕ be a function of t only the two corresponding solutions are:

$$(58) \quad \begin{cases} \phi = at + b, \\ \mu = \log \left[\frac{m^2}{2a^2} \operatorname{sech}^2 \left(-\frac{m}{2} \log \rho + n \right) \right], \\ \lambda = \frac{m^2}{2} \log \rho + p; \end{cases}$$

and

$$(59) \quad \begin{cases} \phi = p - \frac{m}{a} \tanh \left(-\frac{m}{2} t + n \right), \\ \mu = \log \left[\frac{2a^2}{m^2} \rho^2 \cosh^2 \left(-\frac{m}{2} t + n \right) \right], \\ \lambda = \frac{m^2\rho^2}{4} + \log \left[\rho^2 \cosh^4 \left(-\frac{m}{2} t + n \right) \right] + q. \end{cases}$$

Case (iii).

$$\phi \neq 0, \quad \psi = 0.$$

(iii A). *A null electromagnetic field.* If ϕ and ψ be both functions of $(\rho \pm t)$ equation (15) is satisfied when each is a function of $(\rho - t)$ or of $(\rho + t)$. Here we take them as functions of $(\rho - t)$. The case when they are functions of $(\rho + t)$ can be dealt with similarly.

It can easily be seen that (16) and (17) lead to a single equation, viz.

$$(60) \quad \mu_1 + \mu_4 = \frac{1}{\rho}.$$

From (12) we find

$$(61) \quad \mu_{11} - \mu_{44} + \frac{\mu_1}{\rho} = 0.$$

Solving these two equations for μ we find that

$$(62) \quad \mu = \log \left(\frac{\rho}{a} \right).$$

Substituting for μ , ϕ and ψ in (13) and (14) we find that

$$(63) \quad \lambda_1 = \frac{1}{2\rho} + \frac{2}{a} \phi'^2 + 2a\psi'^2, \quad \lambda_4 = -\frac{2}{a} \phi'^2 - 2a\psi'^2,$$

and hence

$$(64) \quad \lambda = \frac{1}{2} \log \left(\frac{\rho}{b} \right) + 2 \int \left(\frac{\phi'^2}{a} + a\psi'^2 \right) du,$$

b being an arbitrary constant. In the above an overhead dash denotes differentiation with respect to $u = (\rho - t)$.

The surfaces $\phi = \text{const.}$ and $\psi = \text{const.}$ are null surfaces and the solution represents a null electromagnetic field *in vacuo*. The metric (1) for this solution is plane symmetric in the sense of Taub (1951) and consequently it represents plane electromagnetic waves.

(iii B). *Further solutions.* If we now assume that each of ϕ and ψ depends upon one variable only it follows from (15) that they should be functions not of the same variable. For if they were functions of the same variable, say ρ , (15) implies

$$\phi_1 \psi_1 = 0,$$

and hence either $\phi_1 = 0$ or $\psi_1 = 0$. In either case we are led to one or the other of the cases (i C) and (ii B) already discussed. A similar conclusion can be drawn if both are functions of t only. When ϕ and ψ are functions not of the same variable two possibilities arise:

(A) ϕ is a function of ρ and ψ a function of t .

(B) ϕ is a function of t and ψ a function of ρ .

(A). In this case $\phi_4 = \psi_1 = 0$. Equation (15) is identically satisfied and from equations (16) and (17) we get the first integrals

$$\mu = -\log \phi_1 + \log \rho + F(t),$$

$$\mu = \log \psi_4 + G(\rho),$$

where F and G are arbitrary functions of t and ρ respectively. From the above two expressions for μ it follows that

$$G(\rho) = -\log \phi_1 + \log \rho, \quad F(t) = \log \psi_4,$$

and hence

$$(65) \quad \mu = \log \left(\frac{\rho \psi_4}{\phi_1} \right).$$

Substituting for μ from (65) in (12) we find

$$(66) \quad -\frac{d^2}{d\rho^2} \log \left(\frac{\phi_1}{\rho} \right) - \frac{1}{\rho} \frac{d}{d\rho} \log \left(\frac{\phi_1}{\rho} \right) - \frac{d^2}{dt^2} \log \psi_4 = \frac{2\phi_1 \psi_4}{\rho}.$$

Since ϕ and ψ are functions of ρ and t respectively it follows that (66) will hold only if $\frac{\phi_1}{\rho}$ be a constant or ψ_4 be a constant. If $\frac{\phi_1}{\rho}$ be a constant, say $\phi_1 = a\rho$, a being an arbitrary constant we get

$$(67) \quad \phi = \frac{1}{2}a\rho^2 + b,$$

and (66) reduces to

$$\frac{d^2}{dt^2} \log \psi_4 = -2a\psi_4,$$

whose general solution is

$$(68) \quad \psi = p - \frac{m}{2a} \tanh \left(-\frac{m}{2}t + n \right).$$

From (65), (67) and (68) we have

$$(69) \quad \mu = \log \left[\frac{m^2}{4a^2} \operatorname{sech}^2 \left(-\frac{m}{2}t + n \right) \right],$$

and finally from (13), (14), (67), (68) and (69) we obtain

$$(70) \quad \lambda = \frac{m^2 \rho^2}{4} + q.$$

If, however, ψ_4 be a constant, say $\psi_4 = a$, the corresponding solution is easily found to be

$$(71) \quad \begin{cases} \psi = at + b, \quad \phi = p - \frac{m}{2a} \tanh \left(-\frac{m}{2} \log \rho + n \right), \\ \mu = \log \left[\frac{4a^2}{m^2} \rho^2 \cosh^2 \left(-\frac{m}{2} \log \rho + n \right) \right], \\ \lambda = \frac{m^2}{2} \log \rho + \log \left[\rho^2 \cosh^4 \left(-\frac{m}{2} \log \rho + n \right) \right] + q. \end{cases}$$

(B). In this case $\phi_1 = \psi_4 = 0$. Equation (15) is identically satisfied and

from the other equations we find, as in the case (A), the following two solutions:

$$(72) \quad \begin{cases} \psi = \frac{1}{2}a\rho^2 + b, \quad \phi = p - \frac{m}{2a} \tanh\left(-\frac{m}{2}t + n\right), \\ \mu = \log\left[\frac{4a^2}{m^2}\rho^2 \cosh^2\left(-\frac{m}{2}t + n\right)\right], \\ \lambda = \frac{m^2\rho^2}{4} + \log\left[\rho^2 \cosh^4\left(-\frac{m}{2}t + n\right)\right] + q; \end{cases}$$

and

$$(73) \quad \begin{cases} \psi = p - \frac{m}{2a} \tanh\left(-\frac{m}{2}\log\rho + n\right), \quad \phi = at + b, \\ \mu = \log\left[\frac{m^2}{4a^2} \operatorname{sech}^2\left(-\frac{m}{2}\log\rho + n\right)\right], \\ \lambda = \frac{m^2}{2} \log\rho + q. \end{cases}$$

4. APPLICATIONS OF BONNOR'S THEOREM

Bonnor (1954) established the theorem:

If g_{ij} , F_{ij} are tensors satisfying (2), (3), (4) and (5) then g_{ij} , ${}^*F_{ij}$ also satisfy them where

$$(74) \quad {}^*F^{ij} = \frac{1}{2}\eta^{ijkl}F_{kl} = -\frac{1}{2}\epsilon_{ijkl}(-g)^{-\frac{1}{2}}F_{kl},$$

ϵ_{ijkl} being the usual permutation symbol.

It can now easily be shown that the solutions obtained in cases (i C) and (ii B) are not all independent in the sense that knowing one of these solutions we can determine another with the help of this theorem. For example, consider the solution given by (43). The nonvanishing components of the electromagnetic field tensor corresponding to this solution are

$$(75) \quad F_{34} = -F_{43} = \frac{m^2}{2a\sqrt{8\pi}} \operatorname{sech}^2\left(-\frac{m}{2}t + n\right).$$

Hence in view of (74) we have

$$(76) \quad {}^*F^{12} = -\frac{1}{2}(\epsilon_{1234}F_{34} + F_{43}\epsilon_{1243})(-g)^{-\frac{1}{2}}.$$

Substituting for the various quantities involved we find that

$${}^*F^{12} = -\frac{a\rho}{\sqrt{8\pi}},$$

i.e.

$$(77) \quad \phi_1 = a\rho.$$

Hence another solution will be given by

$$(78) \quad \begin{cases} \phi = \frac{1}{2}a\rho^2 + b, \\ \mu = \log \left[\frac{m^2}{2a^2} \operatorname{sech}^2 \left(-\frac{m}{2}t + n \right) \right], \quad \lambda = \frac{m^2\rho^2}{4} + g, \end{cases}$$

which is the same as that given by (57).

In this way every pair of two solutions, one from case (i C) and the other from case (ii B), which have the same g_{ij} and differ only in having ϕ or ψ can be shown to be derivable from one another in view of Bonnor's theorem.

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REFERENCES

- Bonnor, W. B. (1954). *Proc. phys. Soc.*, A **67**, 225.
 ——— (1957). *J. Math. Mech.*, **6**, 203.
 Einstein, A., and Rosen, N. (1937). *J. Franklin Inst.*, **223**, 43.
 Levi-Civita, T. (1919). *R. C. Accad. Lincei*, **28** i (3), 101.
 Majumdar, S. D. (1947). *Phys. Rev.*, **72**, 390.
 Pirani, F. A. E. (1957). *Ibid.*, **105**, 1089.
 Rosen, N. (1954). Some cylindrical gravitational waves. *Bull. Res. Coun. Israel*, **3**, 328.
 ——— (1955). Gravitational Waves: Jubilee of Relativity Theory, Bern, 171.
 ——— (1958). *Phys. Rev.*, **110**, 291.
 Taub, A. H. (1951). *Ann. Math.*, **53**, 472.
 Weber, J., and Wheeler, J. A. (1957). *Rev. mod. Phys.*, **29**, 509.
 Weyl, H. (1917). *Ann. Phys. Lpz.*, **54**, 117.